

10.13 DIMENSIONAL REGULARIZATION AND THE MS SCHEME

THE CORE MECHANISM OF DIMENSIONAL REGULARIZATION

In dimensional regularization (dim reg), we analytically continue the spacetime dimension from $d = 4$ to $d = 4 - \epsilon$ (some books use $d = 4 - 2\epsilon$; it is a matter of convention). The fundamental quantity governing divergences is the Euler Gamma function, as seen in the master formula for Euclidean integrals:

$$\int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 + \Delta)^n} = \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(n - \frac{d}{2})}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n-d/2} \quad (10.13.1)$$

The Gamma function $\Gamma(z)$ has simple poles at non-positive integers $z = 0, -1, -2, \dots$. Therefore, divergences (as poles in ϵ) appear when the argument $n - d/2$ approaches one of these integers. For one-loop integrals of this form the Gamma-function poles are the *only* source of ultraviolet divergences: if $\Gamma(n - d/2)$ is finite, the integral is finite in this scheme.

ODD VERSUS EVEN DIMENSIONS

The behavior of loop integrals differs between odd and even dimensions. For a one-loop diagram ($L = 1$) with I internal scalar propagators and no derivative couplings, the superficial degree of divergence is

$$\omega = dL - 2I = d - 2I.$$

Look at its parity:

- If d is even, $\omega = (\text{even}) - 2I$ is even, so $\omega = 0$ (a logarithmic divergence) is possible.
- If d is odd, $\omega = (\text{odd}) - 2I$ is *always* odd.

So, by power counting, a one-loop diagram in an odd dimension cannot be logarithmically divergent. This explains the “special” nature of odd dimensions in dim reg:

- **Even dimensions** ($d = 2, 4, 6, \dots$): $d/2$ is an integer. One-loop diagrams can have $\omega = 0, 2, 4, \dots$. The Gamma-function argument $n - d/2$ can hit a pole, producing $1/\epsilon$ divergences for *both* power-law and logarithmic cases.

- **Odd dimensions** ($d = 3, 5, 7, \dots$): $d/2$ is a half-integer. One-loop diagrams can only have $\omega = 1, 3, 5, \dots$. The Gamma-function argument $n - d/2$ is *always* a half-integer, which is *never* a pole. Thus **all one-loop diagrams in odd dimensions are finite in dim reg**. But we know a divergence is a divergence; it does not disappear. Dim reg does something special, explained below.

Clarification: this only applies at one loop. In odd dimensions, logarithmic divergences appear at even loop orders, where $\omega = Ld - 2I$ can vanish. In ϕ^6 theory in $d = 3$, the one-loop diagrams are finite, but at two loops $\omega = 2d - 2I = 0$ for $I = 3$, and dim reg shows the corresponding $1/\epsilon$ pole. This is why renormalization is still needed in odd dimensions.

POWER-LAW VERSUS LOGARITHMIC DIVERGENCES

Dim reg treats the two kinds of divergence very differently.

- **Scaleless integrals**, which have no mass parameter at all, like $\int d^d k (k^2)^a$, are analytically continued to zero. This is a built-in feature of the scheme. Introducing the renormalization scale μ ($d^d k \rightarrow \mu^{4-d} d^d k$, to keep couplings dimensionless) does not change this: it cannot generate a mass scale inside the integral.
- **Integrals with a mass scale** $\Delta = m^2$ keep only the part proportional to powers of m .

In $d = 4 - \epsilon$:

- **Logarithmic** ($\omega = 0$):

$$I_{\log} = \int \frac{d^d k}{(k^2 + \Delta)^2} \propto \Gamma(2 - d/2) = \Gamma(\epsilon/2) = \frac{2}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon).$$
This is a “pure” $1/\epsilon$ pole.
- **Power law** ($\omega = 2$): $I_{\text{quad}} = \int \frac{d^d k}{k^2 + \Delta} \propto \Gamma(1 - d/2) = \Gamma(-1 + \epsilon/2) \sim m^2 \times \frac{1}{\epsilon}$.
The m^2 comes from the factor $(1/\Delta)^{n-d/2}$ in Eq. (10.13.1), which becomes $\Delta^{d/2-1} \approx \Delta = m^2$.

What does the m^2/ϵ pole correspond to? Compare a Euclidean momentum cutoff Λ :

$$\int^\Lambda \frac{d^4 k}{(2\pi)^4} \frac{1}{k^2 + m^2} = \frac{1}{16\pi^2} \left[\Lambda^2 - m^2 \ln \frac{\Lambda^2}{m^2} + \mathcal{O}\left(\frac{m^4}{\Lambda^2}\right) \right] \quad (\text{cutoff}),$$

$$\int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 + m^2} = -\frac{m^2}{16\pi^2} \left[\frac{2}{\epsilon} + \dots \right] \quad (\text{dim reg}).$$

The quadratically divergent integral contains a subleading *logarithmic* divergence, $-m^2 \ln \Lambda^2$, and that is what the pole is: $\ln \Lambda^2 \leftrightarrow 2/\epsilon$. The pure power Λ^2 has no counterpart in dim reg; it is simply dropped. So it is tempting to say that in even dimensions dim reg “tags” the pole with the

mass dimension of the power-law divergence, but more precisely, **every** $1/\epsilon$ **pole is a logarithmic divergence**, multiplied by whatever powers of the masses the dimensions require. In odd dimensions there is no logarithm at one loop, and a power-divergent integral is evaluated to a finite multiple of m^ω .

Dim reg is subtle: it discards every pure power-law divergence. For a scaleful integral (with masses in Δ) what is left of a divergence of degree ω is m^ω times a $1/\epsilon$ pole (a logarithm) in even dimensions, or times a finite number in odd dimensions. For a scaleless integral nothing is left. This automatic subtraction of pure power divergences is both a feature (computational simplicity) and a limitation (it hides the sensitivity of the results to high scales).

1965 THE $\overline{\text{MS}}$ SCHEME: WHAT ARE WE REALLY DOING?

1966 In the $\overline{\text{MS}}$ (modified minimal subtraction) scheme, we:

- 1967 1. Subtract **only** the $1/\epsilon$ poles (together with the constants $\gamma_E - \ln(4\pi)$ that accompany them).
- 1968 2. Do *not* “ignore” power-law divergences. Rather, we **implicitly**
- 1969 **accept dim reg’s evaluation of them**, finite or zero.
- 1970

1971 This works because in a renormalizable QFT all power-law divergences
1972 can be absorbed by the same counterterms that absorb the logarithmic
1973 ones. The $\overline{\text{MS}}$ scheme appears “minimal” only because the regulator has
1974 already done the heavy lifting of removing $\Lambda^2, \Lambda^3, \dots$

Contrast with Pauli–Villars regularization:

$$\Pi_{\text{PV}}(p^2) \sim c_1 \Lambda^2 + c_2 m^2 \ln \left(\frac{\Lambda^2}{\mu^2} \right) + \text{finite}.$$

1975 Here we explicitly see both quadratic (Λ^2) and logarithmic ($\ln \Lambda^2$)
1976 divergences. A counterterm must be “brute-forced” to cancel the Λ^2 term
1977 explicitly. In $\overline{\text{MS}}$, the Λ^2 term never appears.

1978 ONLY THE $1/\epsilon$ POLES DRIVE THE RUNNING

The renormalization-group flow, described by the β -function $\beta(g_R) = \mu dg_R/d\mu$, tells us how a coupling g_R must change with the scale μ to keep physical observables independent of our choice of μ . The bare coupling g_0 is independent of μ . In $d = 4 - \epsilon$ dimensions,

$$g_0 = \mu^\epsilon \left(g_R + \frac{a_1(g_R)}{\epsilon} + \dots \right).$$

Setting $\mu dg_0/d\mu = 0$ and keeping the ϵ^0 terms gives

$$\beta(g_R) = \left(g_R \frac{d}{dg_R} - 1 \right) a_1(g_R),$$

1979 so β is determined *entirely* by a_1 , the residue of the $1/\epsilon$ pole. For $\lambda\phi^4/4!$
 1980 theory, $a_1 = 3\lambda^2/16\pi^2$ and $\beta = 3\lambda^2/16\pi^2$.

1981 The same holds for the mass. The m^2/ϵ pole of the quadratically divergent
 1982 tadpole is a genuine $1/\epsilon$ residue, and it drives the running of m^2 : in $\lambda\phi^4/4!$
 1983 theory, $m_0^2 = m^2(1 + \lambda/16\pi^2\epsilon + \dots)$, so that $\mu dm^2/d\mu = \lambda m^2/16\pi^2$ at one loop.
 1984 What does *not* run is the pure power Λ^2 , which dim reg has already
 1985 discarded. So the precise statement is: the running is fixed by the
 1986 logarithmic divergences, including the logarithms that sit inside
 1987 power-divergent integrals with mass-dimension prefactors. The finite
 1988 parts left after subtraction are not μ -independent: they carry $\ln(\mu^2/m^2)$
 1989 and $\ln(\mu^2/p^2)$, and their μ -dependence is exactly what the running of the
 1990 couplings compensates.

1991 MASS INDEPENDENCE, THRESHOLDS AND DECOUPLING

1992 Because $\overline{\text{MS}}$ subtracts only the $1/\epsilon$ poles, and their coefficients are
 1993 polynomials in the masses with mass-independent coefficients, the
 1994 renormalization factors Z (and the counterterm coefficients $\delta_m, \delta_\lambda, \dots$) do
 1995 not depend on the masses. The β -functions of the dimensionless
 1996 couplings are therefore mass independent: the scheme is “blind” to the
 1997 masses of the particles in the loops.

1998 The physical consequence is that $\overline{\text{MS}}$ **does not see particle thresholds**.
 1999 When the scale drops below the mass M of a heavy particle, that particle
 2000 should stop affecting low-energy physics (the Appelquist–Carazzone
 2001 decoupling theorem, 1975). In a physical, momentum-subtraction
 2002 scheme this happens automatically: the heavy particle’s contribution to
 2003 the running of the light couplings switches off for $\mu \ll M$. In $\overline{\text{MS}}$ the heavy
 2004 particle keeps contributing to the β -functions at all μ , and its effect at low
 2005 energy reappears as large logarithms $\ln(M/\mu)$ in the finite parts. One
 2006 restores decoupling by hand: integrate the heavy particle out at $\mu \sim M$
 2007 and match the full theory to an effective field theory without it.

2008 **Example:** in the Standard Model the top quark ($m_t \approx 173$ GeV) should not
 2009 affect the running of the light couplings at $\mu \ll m_t$. In $\overline{\text{MS}}$ this has to be
 2010 implemented through decoupling relations: for instance, the strong
 2011 coupling below m_t is run with five quark flavors after matching at $\mu \sim m_t$.

2012 PHYSICAL IMPLICATIONS AND THE LIMITS OF 2013 MATHEMATICAL ELEGANCE

2014 While dim reg + $\overline{\text{MS}}$ is computationally optimal, it obscures important
 2015 physical puzzles by “hiding” power-law divergences.

2016 THE HIERARCHY PROBLEM

In cutoff regularization, the top-quark loop corrects the Higgs mass by

$$\delta m_H^2 \sim y_t^2 \Lambda^2 + y_t^2 m_t^2 \ln \left(\frac{\Lambda^2}{\mu^2} \right).$$

The quadratic term Λ^2 represents a dramatic fine-tuning problem if $\Lambda \sim M_{\text{Pl}}$. In dim reg,

$$\delta m_H^2 \sim y_t^2 m_t^2 \left(\frac{1}{\epsilon} + \text{finite} \right).$$

2017 The Λ^2 divergence is gone. The fine-tuning problem becomes
2018 mathematically invisible, but physically it remains: if the Higgs couples
2019 to any heavy particle of mass M , dim reg gives a correction proportional
2020 to M^2 , and the sensitivity to the heavy scale is back.

2021 THE COSMOLOGICAL CONSTANT PROBLEM

With a cutoff, the zero-point energy density of a free field is

$$\rho_{\text{vac}} = \frac{1}{2} \int^\Lambda \frac{d^3 k}{(2\pi)^3} \sqrt{\mathbf{k}^2 + m^2} \sim \frac{\Lambda^4}{16\pi^2}.$$

2022 In dim reg, for a massless field this is a scaleless integral, which the rules
2023 define to be **zero**; for a massive field only a term proportional to
2024 $m^4 \ln(m^2/\mu^2)$ survives. This regularization *deletes* the Λ^4 from the
2025 calculation, but it does not remove the problem: the m^4 contributions of
2026 the known particles alone exceed the observed vacuum energy by many
2027 orders of magnitude.

2028 THE LESSON

2029 (As I emphasized in class.) We must distinguish between mathematical
2030 convenience and physical insight. Different (reasonable) regularization
2031 schemes make different problems or properties of the underlying theory
2032 apparent. Dim reg's mathematical elegance comes at the cost of physical
2033 transparency for fine-tuning problems and decoupling phenomena. Dim
2034 reg gives the correct answers for renormalized observable quantities (like
2035 scattering amplitudes) but may obscure the deeper physical questions
2036 about naturalness, UV sensitivity, and decoupling that are made
2037 manifest in cutoff-based schemes. The choice of regularization scheme is
2038 not just computational: it embodies a philosophy about what constitutes
2039 "real" physics. For a bottom-up effective field theorist, a renormalizable
2040 theory is self-consistent to arbitrarily high scales, and won't see any
2041 problem with the underlying theory. For a top-down UV theorist who
2042 wants to understand the ultimate Lagrangian of nature, consistency is
2043 only one aspect; calculability (and hence the predictivity of the bare

parameters) is crucial, as long as one does not want to admit that we live in a special and randomly tuned universe. Misinterpretations of the $\overline{\text{MS}}$ scheme have caused a tremendous amount of confusion in the field, even today. I hope you learned something from this note.

SUMMARY

- **Mathematical advantage:** $\text{dim reg} + \overline{\text{MS}}$ is the most efficient scheme, preserving symmetries and keeping only the universal logarithmic running.
- **Odd dimensions** are a clean laboratory: one-loop finiteness by power counting, and $1/\epsilon$ poles only at even loop orders ($L = 2, 4, \dots$) in scalar theories.
- **Every $1/\epsilon$ pole is a logarithm.** Pure power divergences are discarded, and the m^2/ϵ of a quadratically divergent integral is the logarithm inside it; it drives the running of m^2 .
- **Physical limitation:** the scheme makes fine-tuning problems (like the hierarchy problem) mathematically invisible, sets scaleless power divergences (like the massless vacuum energy) to zero, and, being mass independent, does not decouple heavy particles by itself.